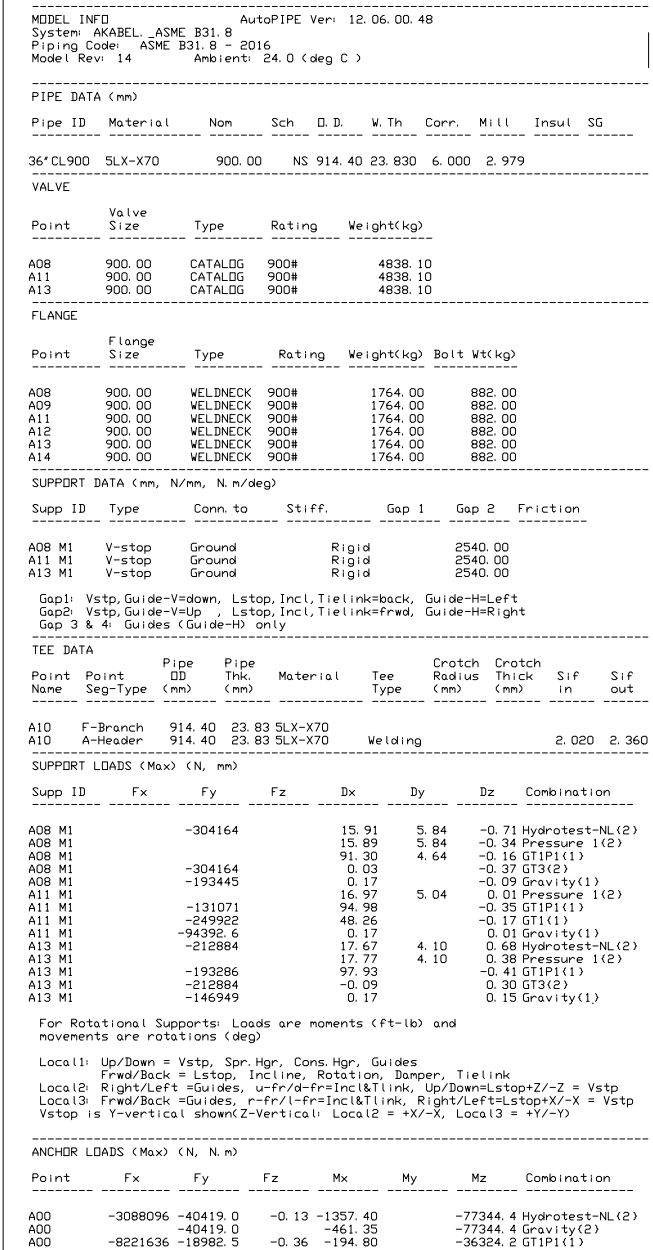
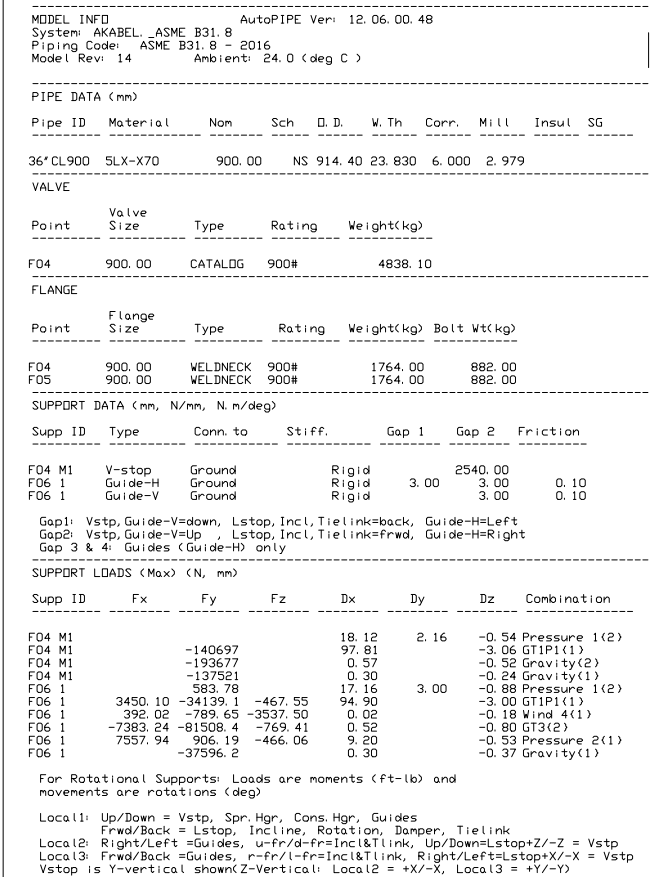
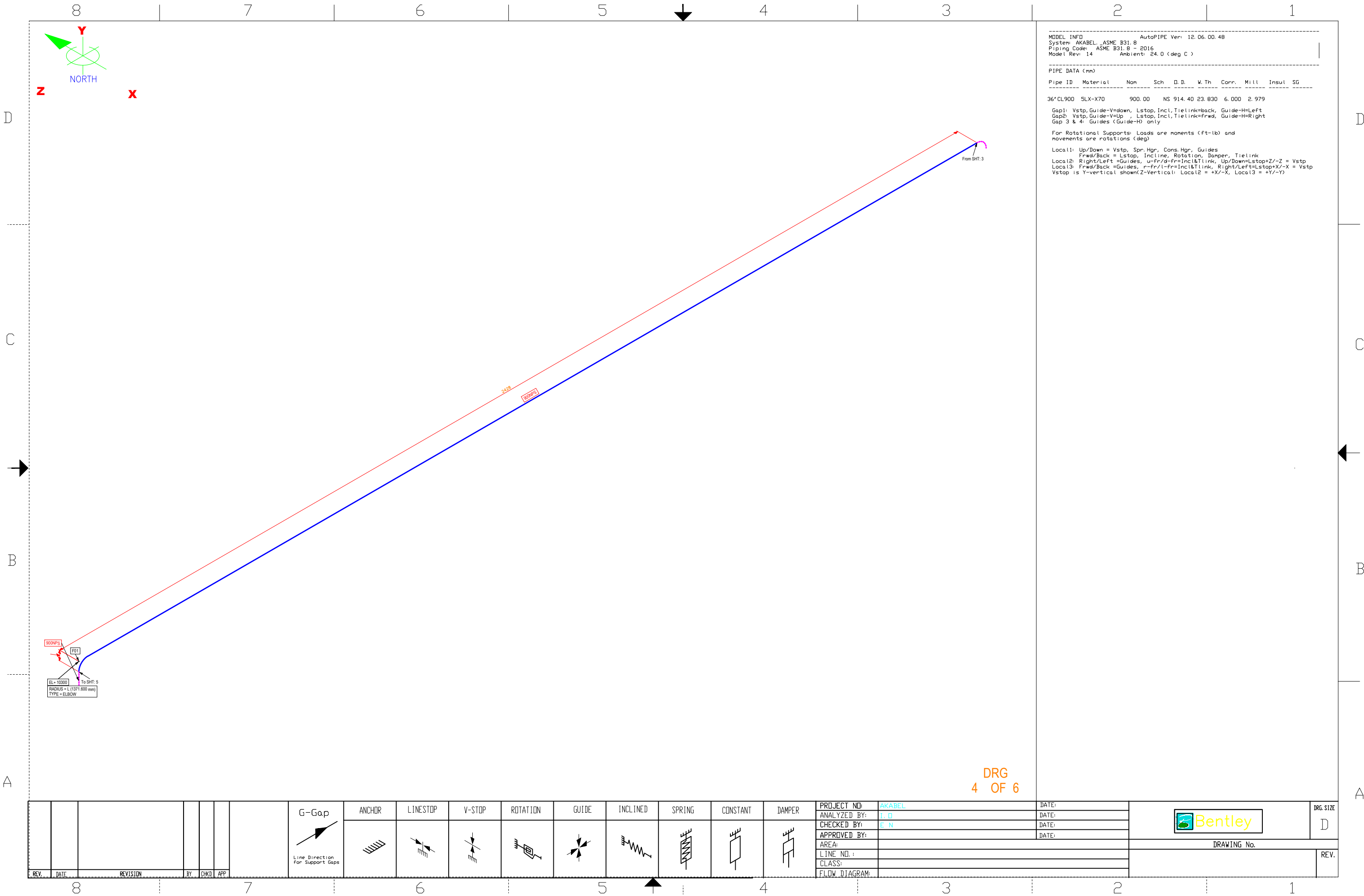


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MODEL INFO
System: AKABEL, ASME B31.8
Piping Code: ASME B31.8 - 2016
Model Revi: 14
AutoPIPE Ver: 12.06.00.48
Ambient: 24.0 (deg C)

PIPE DATA (mm)

Pipe ID	Material	Nom	Sch	D.D.	W.Th	Corr.	Mill	Insul	SG
36'CL900	SLX-X70	900.00	NS	914.40	23.830	6.000	2.979		

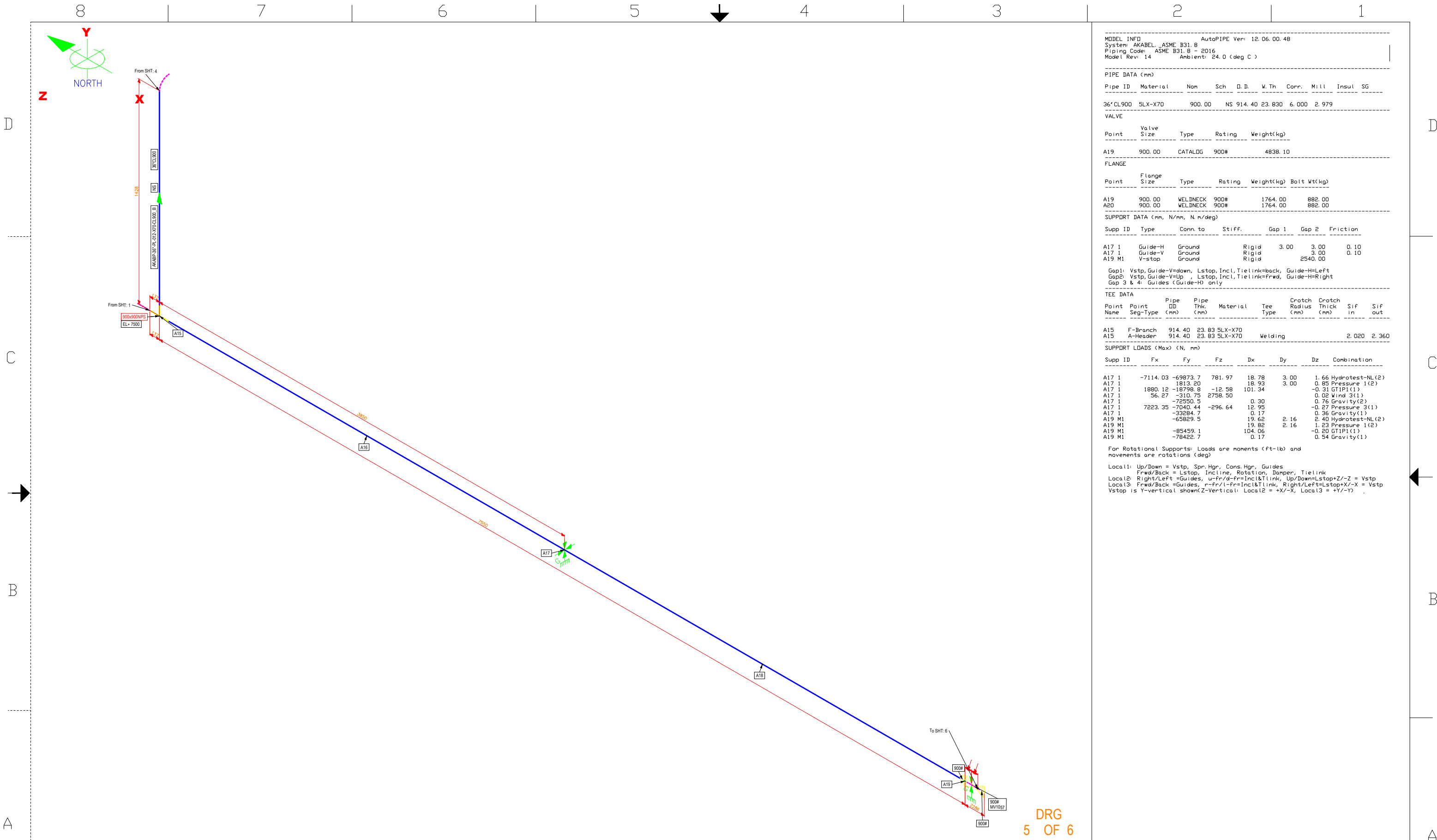
Gap1: Vstop, Guide-V=down, Lstop, Incl, Tielink=back, Guide-H=Left
Gap2: Vstop, Guide-V=Up, Lstop, Incl, Tielink=frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

For Rotational Supports: Loads are moments (ft-lb) and movements are rotations (deg)

Local1: Up/Down = Vstop, Spr.Hgr, Cons.Hgr, Guides
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left = Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstop
Local3: Frwd/Back = Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstop
Vstop is Y-vertical shown(Z-vertical) Local2 = +X/-X, Local3 = +Y/-Y

							G-Gap  Line Direction For Support Gaps	ANCHOR 	LINESTOP 	V-STOP 	ROTATION 	GUIDE 	INCLINED 	SPRING 	CONSTANT 	DAMPER 	PROJECT NO: ANALYZED BY: CHECKED BY: APPROVED BY: AREA: LINE NO. : CLASS: FLOW DIAGRAM:	AKABEL T. B E. N 	DATE: DATE: DATE: DATE: 	Bentley	DRG. SIZE D
																DRAWING No.		REV.			
REV	DATE	REVISION			BY	CHKD	APP														

DRG
4 OF 6



MODEL INFO
System: AKABEL, _ASME B31.8
Piping Code: ASME B31.8 - 2016
Model Rev: 14 Ambient: 24.0 (deg C)

PIPE DATA (mm)									
Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
36"CL900	SLX-X70	900.00	NS	914.40	23.830	6.000	2.979		

VALVE				
Point	Valve Size	Type	Rating	Weight(kg)
A19	900.00	CATALOG	900#	4838.10

FLANGE					
Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)
A19	900.00	WELDNECK	900#	1764.00	882.00
A20	900.00	WELDNECK	900#	1764.00	882.00

SUPPORT DATA (mm, N/mm, N, n/deg)						
Supp ID	Type	Conn. to	Stiff.	Gap 1	Gap 2	Friction
A17 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
A17 1	Guide-V	Ground	Rigid		3.00	0.10
A19 M1	V-stop	Ground	Rigid		2540.00	

Gap1: Vstop, Guide-V=down, Lstop, Incl, Tielink=back, Guide-H=Left
Gap2: Vstop, Guide-V=Up, Lstop, Incl, Tielink=frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

TEE DATA									
Point Name	Point Seg-Type	Pipe OD (mm)	Pipe Thk (mm)	Material	Tee Type	Crotch Radius (mm)	Crotch Thick (mm)	Sif in	Sif out
A15	F-Branch	914.40	23.83	SLX-X70					
A15	A-Header	914.40	23.83	SLX-X70	Welding			2.020	2.360

SUPPORT LOADS (Max) (N, mm)							
Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination
A17 1	-7114.03	-69873.7	781.97	18.78	3.00	1.66	Hydrotest-NL(2)
A17 1		1813.20		18.93	3.00	0.85	Pressure 1(2)
A17 1	1880.12	-18798.8	-12.58	101.34		-0.31	GTIP1(1)
A17 1	56.27	-310.75	2758.50			0.02	Wind 3(1)
A17 1		-72550.5		0.30		0.76	Gravity(2)
A17 1	7223.35	-7040.44	-296.64	12.95		-0.27	Pressure 3(1)
A17 1		-33284.7		0.17		0.36	Gravity(1)
A19 M1		-65829.5		19.62	2.16	2.40	Hydrotest-NL(2)
A19 M1				19.82	2.16	1.23	Pressure 1(2)
A19 M1	-85459.1			104.06		-0.20	GTIP1(1)
A19 M1		-78422.7		0.17		0.54	Gravity(1)

For Rotational Supports: Loads are moments (ft-lb) and movements are rotations (deg)
Local1: Up/Down = Vstop, Spr, Hgr, Cons, Hgr, Guides
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left = Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstop
Local3: Frwd/Back = Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstop
Vstop is Y-vertical shown(Z=Vertical: Local2 = +X/-X, Local3 = +Y/-Y)

REV.	DATE	REVISION	BY	CHKD	APP

G-Gap

ANCHOR

LINESTOP

V-STOP

ROTATION

GUIDE

INCLINED

SPRING

CONSTANT

DAMPER

PROJECT NO:

ANALYZED BY:

CHECKED BY:

APPROVED BY:

AREA:

LINE NO.:

CLASS:

FLOW DIAGRAM:

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DRAWING No.

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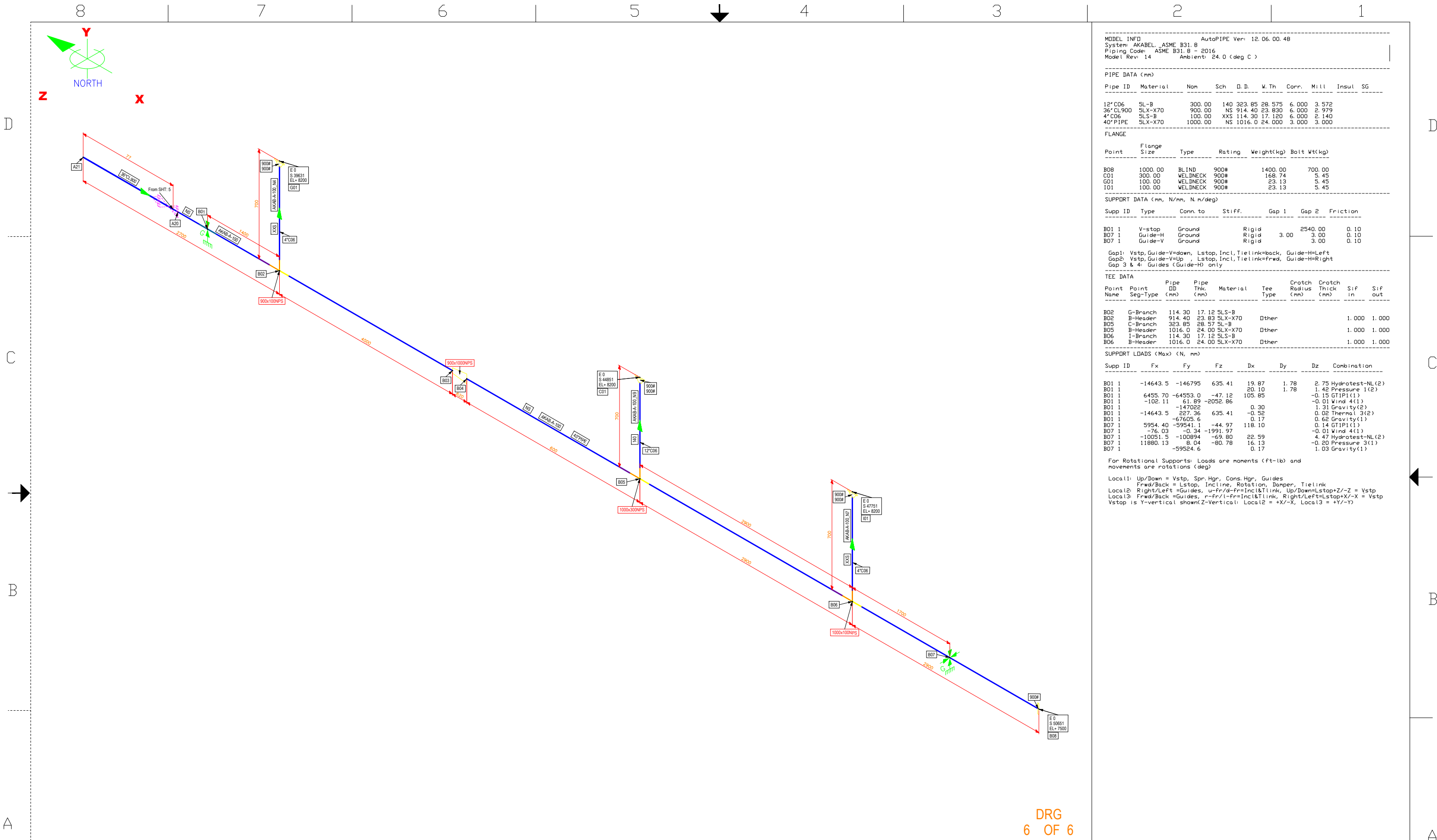
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MODEL INFO
System: AKABEL, _ASME B31.8
Piping Code: ASME B31.8 - 2016
Model Rev: 14
AutoPIPE Ver: 12.06.00.48
Ambient: 24.0 (deg C)

PIPE DATA (mm)									
Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
12" C06	SL-B	300.00	140	323.85	28.575	6.000	3.572		
36" CL900	SLX-X70	900.00	NS	914.40	23.830	6.000	2.979		
4" C06	SL-B	100.00	XXS	114.30	17.120	6.000	2.140		
40" PIPE	SLX-X70	1000.00	NS	1016.0	24.000	3.000	3.000		

FLANGE						
Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)	
B08	1000.00	BLIND	900#	1400.00	700.00	
C01	300.00	WELDNECK	900#	168.74	5.45	
I01	100.00	WELDNECK	900#	23.13	5.45	

SUPPORT DATA (mm, N/mm, N.m/deg)						
Supp ID	Type	Conn. to	Stiff.	Gap 1	Gap 2	Friction
B01 1	V-stop	Ground	Rigid		2540.00	0.10
B07 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
B07 1	Guide-V	Ground	Rigid		3.00	0.10

Gap1: Vstop, Guide-Vdown, Lstop, Incl, Tielink=back, Guide-H=Left
Gap2: Vstop, Guide-Vup, Lstop, Incl, Tielink=frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

TEE DATA									
Point Name	Point Seg-Type	Pipe DD (mm)	Pipe Thk (mm)	Material	Tee Type	Crutch Radius (mm)	Crutch Thick (mm)	Sif in	Sif out
B02	G-Branch	114.30	17.12	SL-B					
B02	B-Header	914.40	23.83	SLX-X70	Other			1.000	1.000
B05	C-Branch	323.85	28.57	SL-B					
B05	B-Header	1016.0	24.00	SLX-X70	Other			1.000	1.000
B06	I-Branch	114.30	17.12	SL-B					
B06	B-Header	1016.0	24.00	SLX-X70	Other			1.000	1.000

SUPPORT LOADS (Max) (N, mm)							
Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination
B01 1	-14643.5	-146795	635.41	19.87	1.78	2.75	Hydrotest-NL(2)
B01 1				20.10	1.78	1.42	Pressure 1(2)
B01 1	6455.70	-64553.0	-47.12	105.85		-0.15	GTIP1(1)
B01 1	-102.11	61.89	-2052.86			-0.01	Wind 4(1)
B01 1		-147022		0.30		1.31	Gravity(2)
B01 1	-14643.5	227.36	635.41	-0.52		0.02	Thermal 3(2)
B01 1		-67605.6		0.17		0.62	Gravity(1)
B07 1	5954.40	-59541.1	-44.97	118.10		0.14	GTIP1(1)
B07 1	-76.03	-0.34	-1991.97			-0.01	Wind 4(1)
B07 1	-10051.5	-100894	-69.80	22.59		4.47	Hydrotest-NL(2)
B07 1	11880.13	8.04	-80.78	16.13		-0.20	Pressure 3(1)
B07 1		-59524.6		0.17		1.03	Gravity(1)

For Rotational Supports: Loads are moments (Ft-lb) and movements are rotations (deg)
Local1: Up/Down = Vstop, Spr, Hgr, Cons, Hgr, Guides
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left = Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstop
Local3: Frwd/Back = Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstop
Vstop is Y-vertical shown(Z-Vertical) Local2 = +X/-X, Local3 = +Y/-Y

DRG
6 OF 6

REV.

DATE

REVISION

BY

CHKD

APP

G-Gap

ANCHOR

LINESTOP

V-STOP

ROTATION

GUIDE

INCLINED

SPRING

CONSTANT

DAMPER

Line Direction For Support Gaps

PROJECT NO: AKABEL

ANALYZED BY: T. D

CHECKED BY: E. N

APPROVED BY:

AREA:

LINE NO.:

CLASS:

FLOW DIAGRAM:

DATE:

DATE:

DATE:

DATE:

DRAWING No.

REV.

DRG SIZE

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